Noel Sason

noel_sason@berkeley.edu | Berkeley, CA | github.com/noelsason | noelsason.com

Education

University of California, Berkeley — *B.A. Molecular Biology | B.A. Data Science* Aug 2024 – Dec 2027

GPA: 3.8 / 4.0 | **Relevant Coursework:** CS61B (Data Structures & Algorithms), Math 54 (Linear Algebra), Data 8 (Foundations of Data Science — Statistics & Probability), Data C88C (Computational Structures in Data Science)

Experience

Research Affiliate — *Lawrence Berkeley National Laboratory* Jan 2026 – Present

- Enabled cross-cohort meta-analysis across 80+ disease–microbiome studies by building a DuckDB-backed pipeline over the KG-Microbe knowledge graph for SQL-based traversal of taxa, chemicals (CHEBI), environments (ENVO), and phenotypic traits.
- Accelerated study onboarding for 4 disease cohorts (colorectal cancer, lung cancer, epilepsy, skin wounds) by writing scripts that transform BugSigDB differential-abundance exports into per-study directories of taxa lists keyed to NCBI Taxon IDs.
- Reduced manual data-wrangling time by designing a YAML-driven trait-profile generator that parses results across all study directories and emits wide and long TSV tables (study $\times$ phenotype) for downstream statistical analysis.
- Improved team reproducibility, as measured by zero environment-setup issues across contributors, by standardizing workflows with Justfile task runners, uv-managed Python environments, and PR-based collaboration on GitHub.

Projects

Canvascope & Lectra — canvascope.org Founder / CEO / Lead Developer

- Grew Canvascope to 46 active users across 10+ university Canvas domains by founding a privacy-first AI study assistant (Chrome extension, Manifest V3) with GCP OAuth authentication, local-first indexing, and sub-100ms semantic search.
- Achieved citation-grounded AI responses, as measured by every answer linking to its source document, by designing a RAG pipeline that indexes course PDFs, pages, and assignments into vector embeddings and retrieves relevant context at query time.
- Developed Lectra (iOS/iPad), a native study workstation: real-time collaborative Apple Pencil annotation (CRDT op-log convergence over Supabase Realtime) plus DropBridge, a custom high-speed file-transfer protocol, and a sandboxed on-device coding IDE, a from-scratch POSIX shell, real git via isomorphic-git bridged to native Swift, CPython via Pyodide (WASM), an SSH terminal, a Claude-powered agent, and a WebRTC Mac–iPad remote desktop, validated by 24 automated tests.

PersonalGraph MCP — github.com/NoelSason/MCP Creator

- Built a local-first MCP server exposing 16 tools, 5 resources, and 4 prompt templates over a typed personal-knowledge graph (nodes/edges with confidence, source, and visibility metadata), validated end-to-end with a live-stdio-protocol smoke test, and adopted as the active memory backend for a production AI coding assistant session.
- Designed a four-tier privacy model (public/ai_allowed/private/sensitive) with double-gated sensitive-data access, input-boundary secret redaction, and append-only audit logging, then decoupled persistence behind a `GraphStore` interface to allow a documented, code-free migration path from JSON to Neo4j/SQLite/Postgres.

SCI Cause-List Alerts Creator

- Designed and built an end-to-end Next.js/Prisma/Postgres alert pipeline, bot-protected and rate-limited (Cloudflare Turnstile, Upstash Redis), that parses government-published court PDFs and emails subscribers the moment their case is listed, achieving idempotent delivery verified by a Vitest parser suite — built to reach litigants who lack easy access to legal counsel or repeated court-website checks.

CLENS — github.com/NoelSason/clens | DataHacks 2026, Scripps Challenge Track Software Engineer

- Built an iOS app that scores grocery purchases on ocean health by training a binary CoreML classifier on 400+ images to auto-detect receipts vs. product barcodes at 6 fps, using a custom hysteresis state machine to dynamically route the scanning camera UI.
- Generated a live marine stress index by processing Scripps CCE2 mooring datasets over OPeNDAP, applying STL seasonal decomposition and an Isolation Forest to flag anomalies that dynamically penalize high-impact items.
- Enabled on-device footprint scoring by programming a zero-dependency ridge regression model in Python that codegens Swift coefficients to predict CO2, runoff, water, and plastic impact, wired to Apple Vision OCR, a Flask API, and a Supabase database.

Technical Skills & Certifications

Languages:	Python, Java, Swift/SwiftUI, TypeScript, JavaScript, SQL, Bash, HTML5/CSS
ML & Data:	PyTorch, scikit-learn, Pandas, NumPy, DuckDB, MongoDB, ONNX Runtime Web, embeddings & vector search
Quant & Methods:	Linear algebra, statistics & probability, A/B testing, data structures & algorithms
Web & Mobile:	Node.js, Next.js, React, Flask, Fastify, iOS/SwiftUI, Xcode, Chrome Extensions, REST API design
Tools & Platforms:	Git, GitHub Actions, Docker, Linux/Unix, Supabase, Vercel, Stripe, Cloudflare R2, GCP/OAuth
Systems & Homelab:	Raspberry Pi, OpenMediaVault (NAS), Tailscale, Pi-hole, Justfile, uv **Cert:** AHA BLS